

Hearing loss: the condition and its impact

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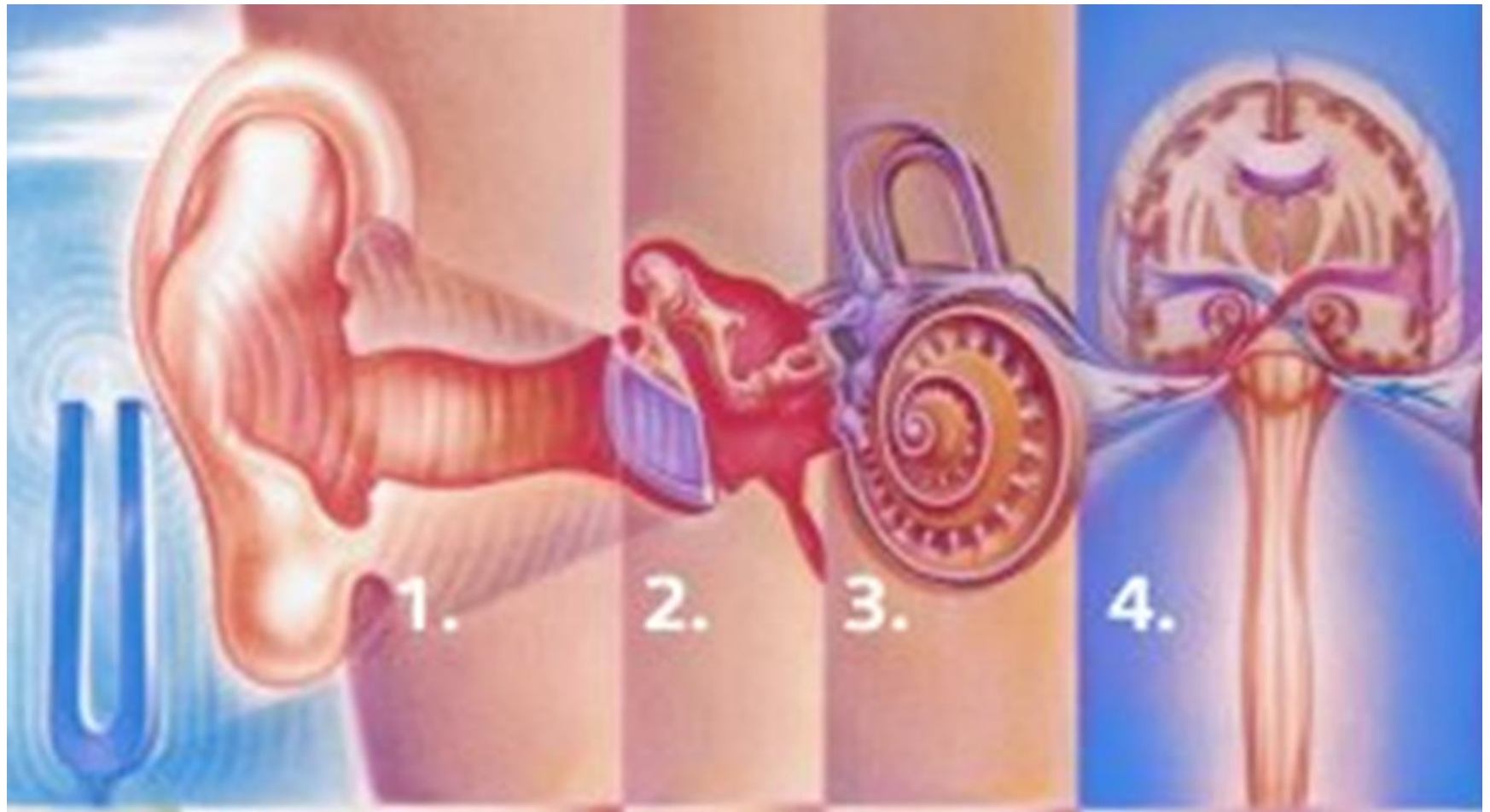
Sound and the ear

- Sound consists of vibrations of air in the form of waves
- These vibrations are converted into electrical signal that are sent to the brain
- In the brain, these signals are translated into meaningful information
- The volume of sound is measured in decibels (dB).

The ear

Consists of three parts:

1. The outer ear
2. The middle ear
3. The inner ear



The outer ear

- Localise sound source
- Collect sound waves and direct them down the canal to the tympanic membrane
- Produces cerumen (wax) assisting in cleaning and lubrication, and also provides some protection from bacteria, fungi and even insects
- Sound is amplified in the ear canal when compressed in the narrow channel.

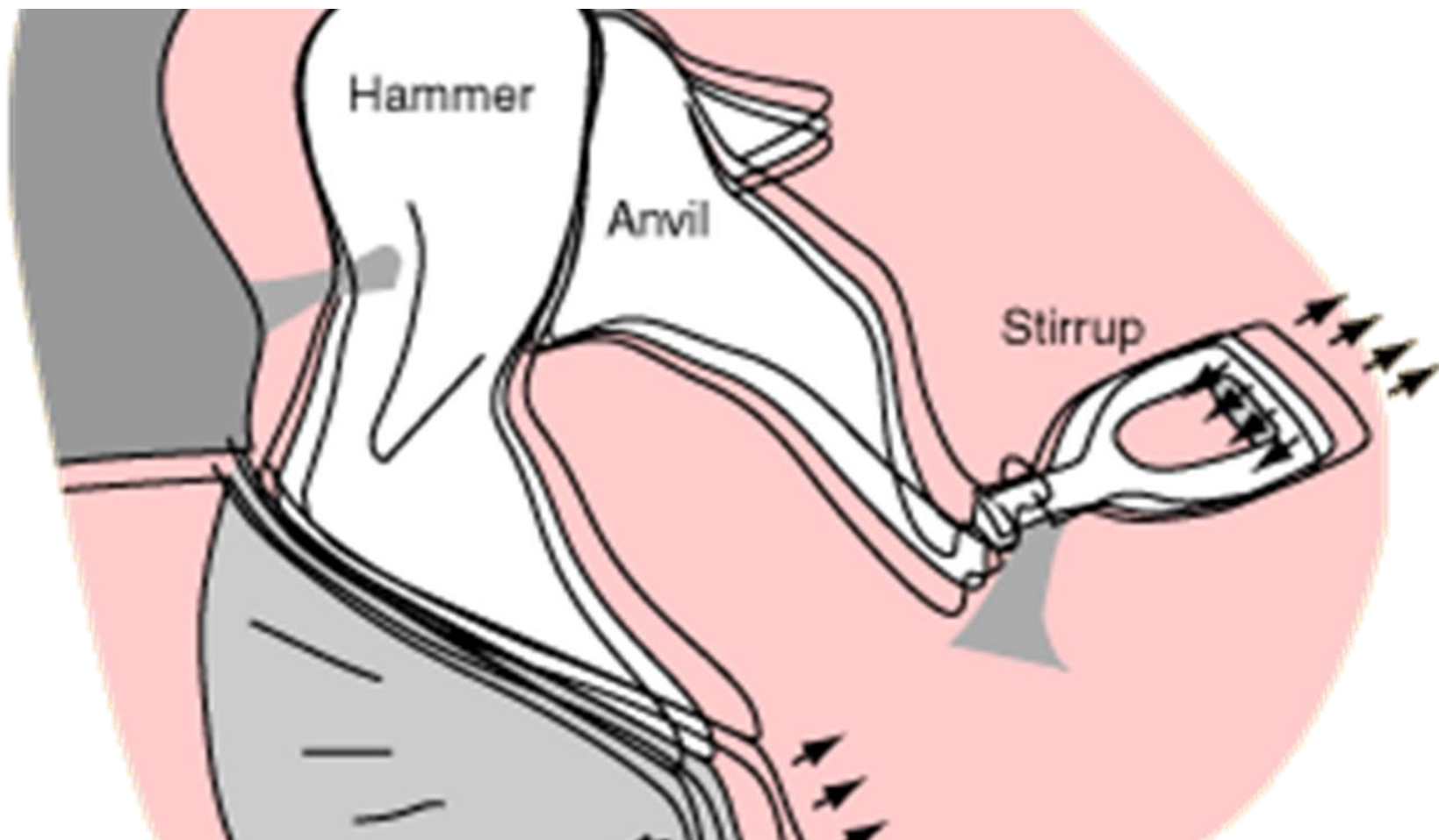
The pinna collects auditory signals.



The middle ear

- Converts sound to a mechanical vibration which transports the sound to the cochlea
- Air pressure applied to the tympanic membrane creates vibrations through the ossicles
- The ossicles vibrate in turn and transfer the sound to the inner ear
- Sound attenuation – to protect the ear from loud sounds.

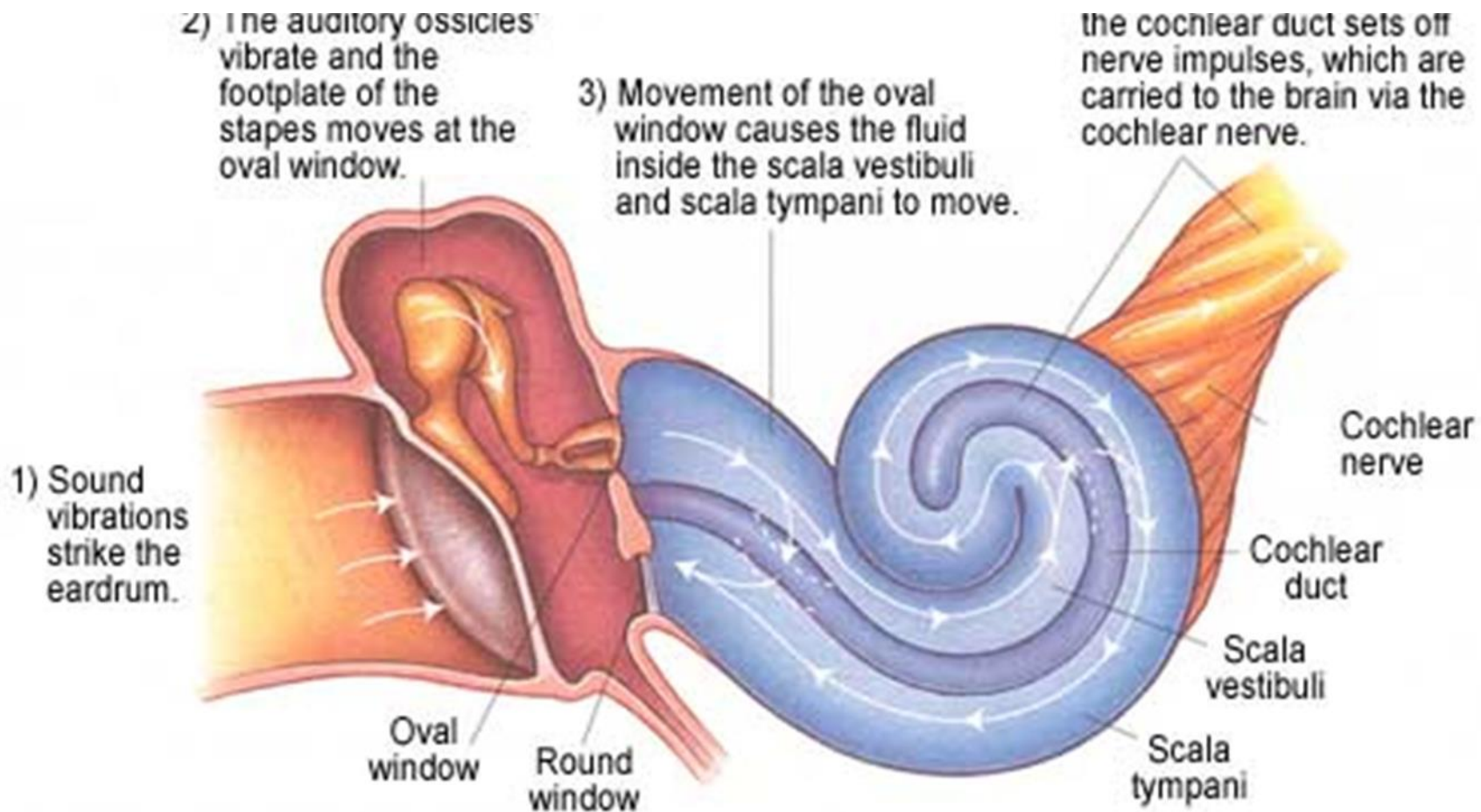
The tympanic membrane vibrations stimulate the malleus which in turn stimulates the incus and the stapes.



The inner ear

- Sound vibrations travel through the oval window
- Oval window is 17x smaller than TM
- Cochlea fluid (perilymph and endolymph) create waves which disturb the basilar membrane
- The inner hair cells in the cochlea attached to the basilar membrane convert the waves into electrical impulses
- Impulses are transmitted to the brain via the auditory nerve.

The hearing system.



What can cause hearing loss?

- The simplest cause is a build-up of wax or other foreign bodies
- A cold or sinus may cause some fullness and reduced hearing
- Flying, going up the mountains or diving due to the difference in pressure between the air in the middle ear and the outside air
- Infection in the middle or outer ear.

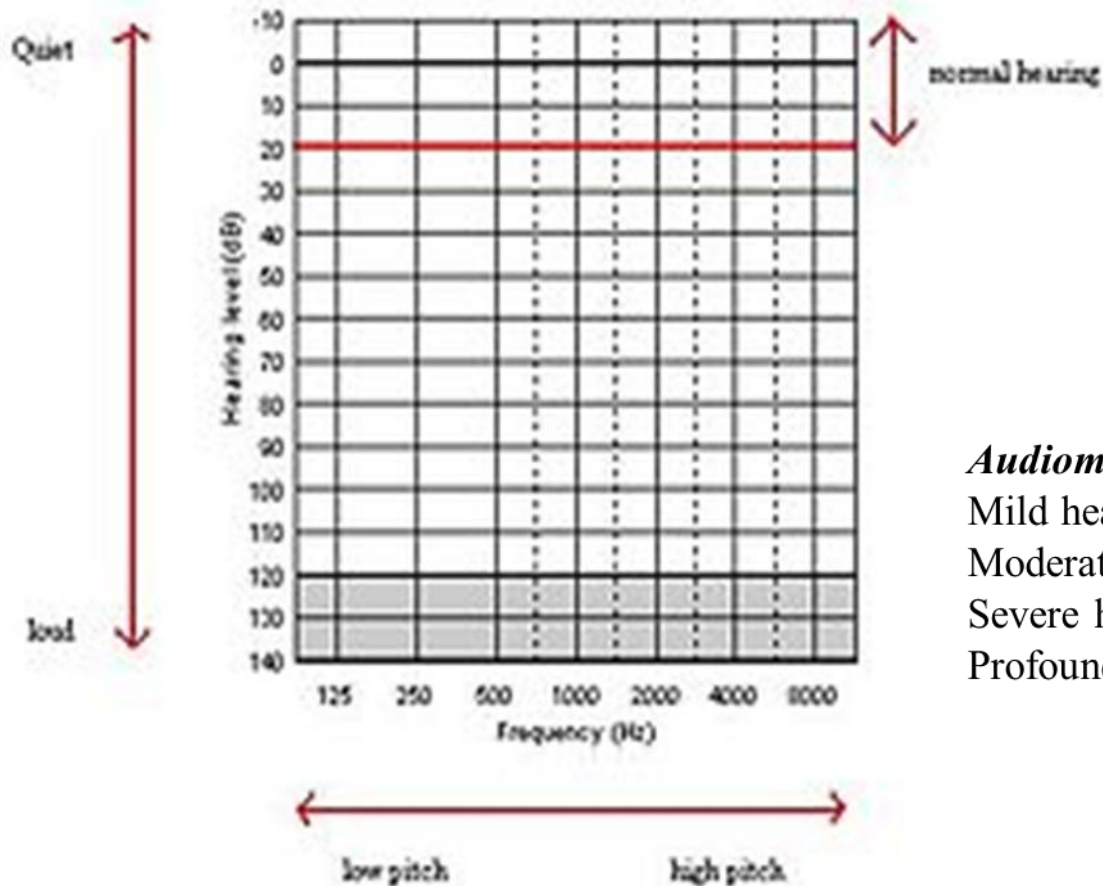
What can cause hearing loss?

- Perforation in the ear drum
- Exposure to loud noise
- Ototoxic drugs
- Head or ear trauma
- Aging
- Congenital or hereditary factors
- Diseases and other causes.

Degrees of hearing loss

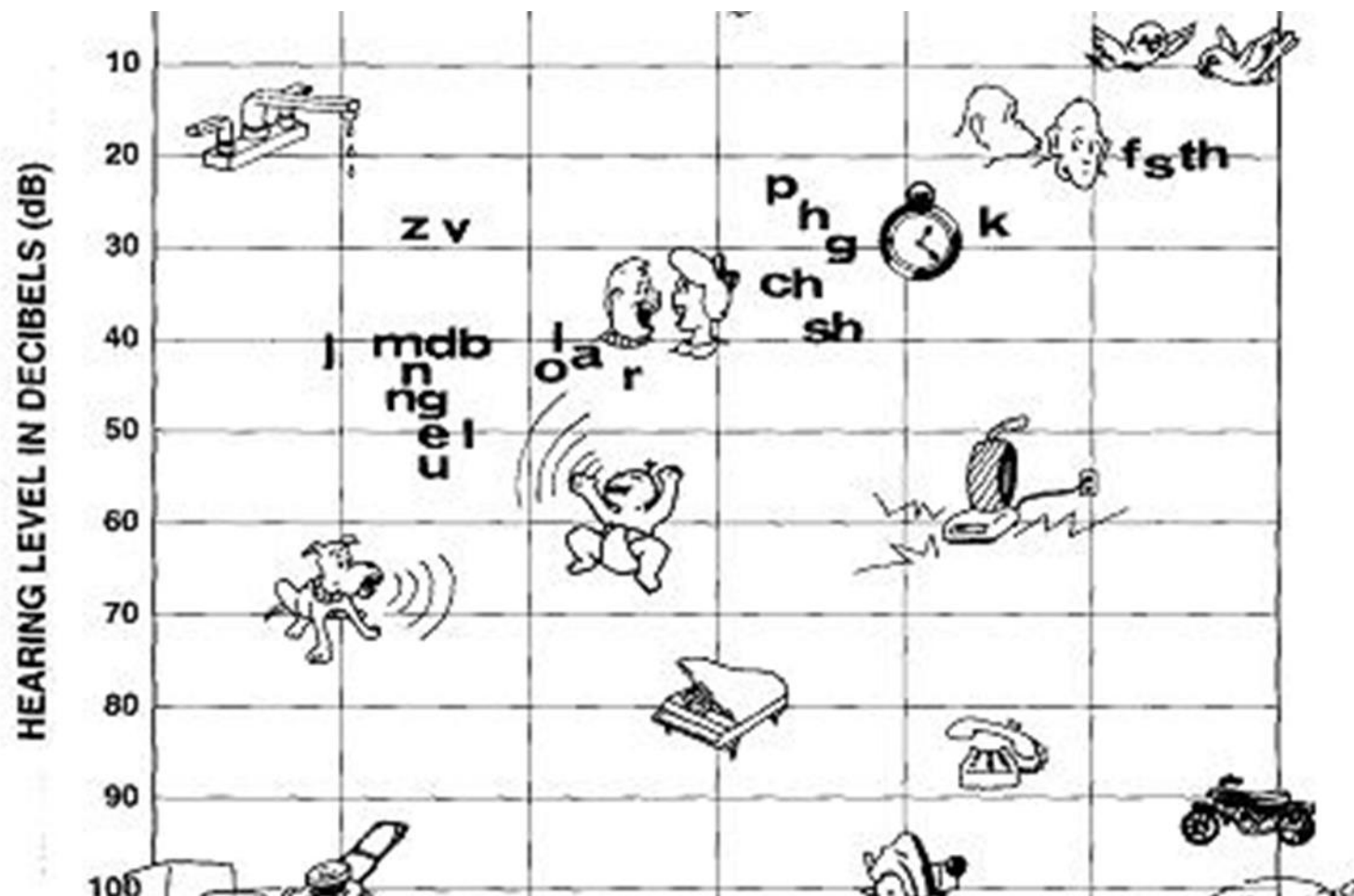
- Results of audiometric evaluation are plotted in a chart – AUDIOGRAM
- Loudness is plotted from top to bottom
- Frequency is plotted from left to right (250 Hz to 8K Hz)
- Hearing loss (HL) is measured in decibels (dB)
- Hearing loss is not measured in percentage.

Audiogram



<i>Audiometric descriptors</i>	<i>HTL (dB)</i>
Mild hearing loss	20-40
Moderate hearing loss	41-70
Severe hearing loss	71-95
Profound hearing loss	In excess of 95

Speech discrimination



Types of hearing loss

- Conductive – involves the outer and middle ear
- Sensorineural – involves the inner ear and auditory nerve
- Mixed – involves both middle and inner ear
- Central – related to the brain rather than the ear.

Conductive hearing loss

- Sound is not conducted efficiently through the ear canal, eardrum or the tiny bones in the middle ear to the inner ear
- Results in reduction of the loudness of sound that is heard
- Ears seem to be full or plugged
- Person may speak softly as they can hear their own voices
- Can be medical or surgically treated.

Sensorineural hearing loss

- Sensory (Cochlea)
- Neural (Retro Cochlear).

Sensory (cochlear) hearing loss

- Most common type of hearing loss
- Permanent hearing loss
- Age related (presbycusis) – main cause
- Damaged hair cells
- May be a result from disturbance of inner ear circulation or increased inner ear pressure or from disturbances of nerve transmission
- Intolerance for loud sounds (recruitment)

Neural (retro cochlear) hearing loss

- Nerve damage
- Tumours on pathway to brain (eg NF2)
- Brain damage
- Poor speech discrimination
- Adaption/tone decay

Central hearing loss

- Auditory Centres of brain are affected by injury, tumour, multiple sclerosis, stroke, hereditary or unknown causes
- Hearing threshold might be normal – CAPD
- Speech perception vs. tonal audibility.

Mixed hearing loss

- Combination of sensorineural and conductive loss
- i.e. presbycusis and middle ear dysfunction
- Late stage of otosclerosis
- Hearing can improve if conductive element is treated.

Associated conditions

- Balance disorder
- Tinnitus
- Depression
- Anxiety
- Dementia

“Adding life to years, not years to life”

ILC-UK Commission on Hearing Loss 2014



Over 10 million people in the UK have hearing loss – accounting for **one in six of the population.**



Yet only **2 million out of 6 million people** who could benefit from hearing aids have access to them.



Of **2 million people** who have hearing aids, **30% do not regularly use them.**



By 2031 it is estimated that **over 14.1 million people** will have hearing loss – accounting for **19.8% of the population.**⁵

5 Action on Hearing Loss (2011), ONS Population Projections: Principal Projection 2012 and author's calculations.

Incidence rises to 1:3 of over 65's

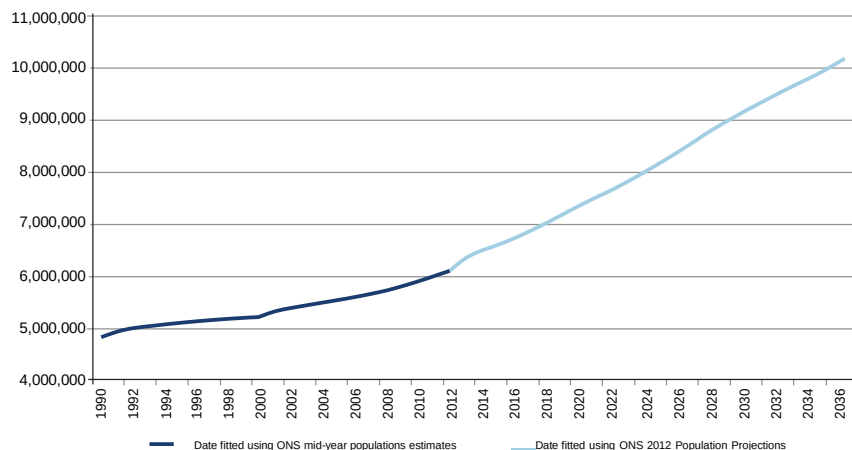
Hearing loss is associated with higher incidence of multiple disability

One of the greatest challenges is recognising hearing loss – and accepting it

Hearing loss

- Burden of disease due to hearing loss is significant, high & rising

Figure 1: Estimates of number of people with hearing loss of at least 35 db across the UK 1990-2037



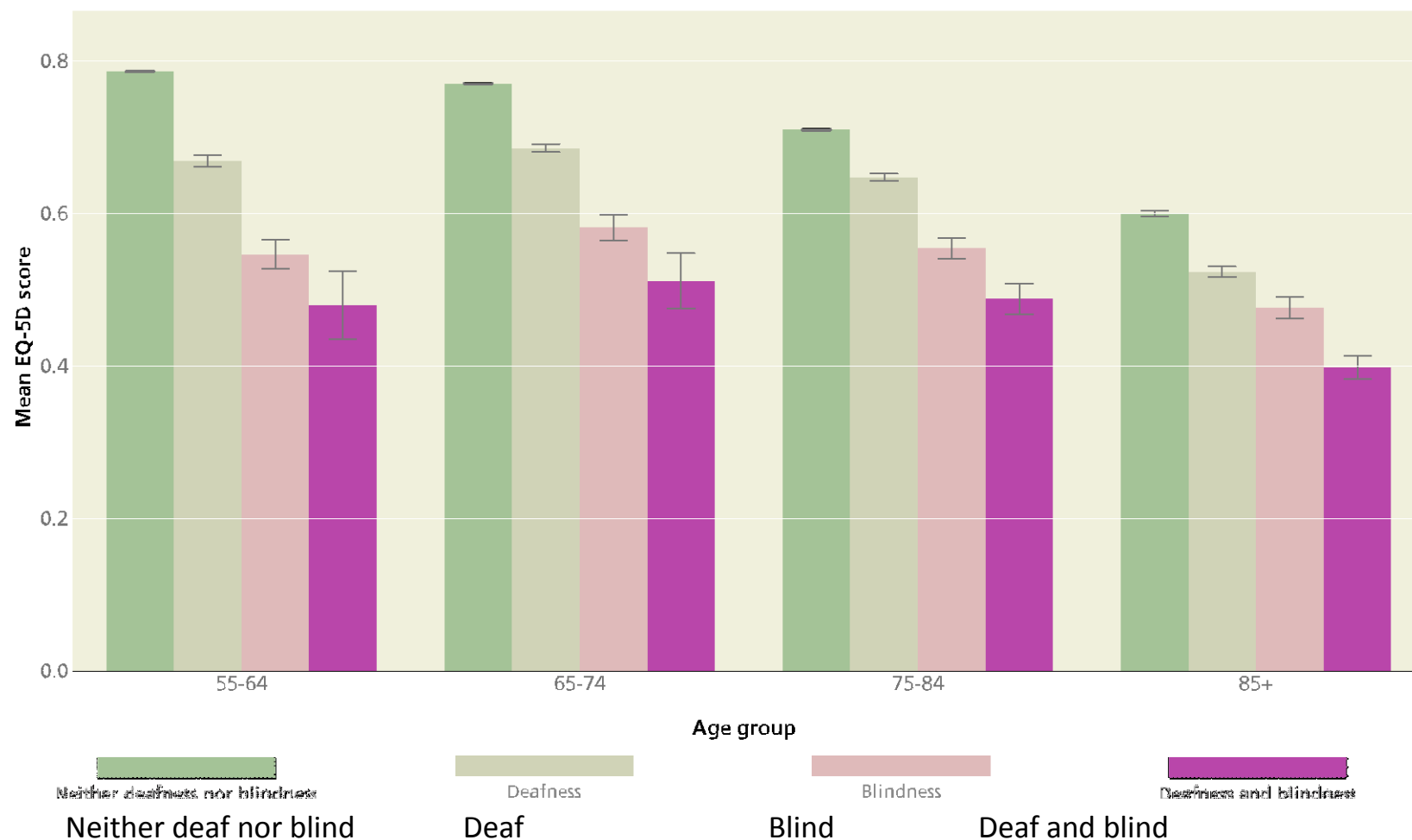
Source: ONS, Forman and Holman (2014) and author's calculations

ILC-UK Hearing Loss 2014

As the population ages, so hearing loss will increase

- Economic cost to UK 2013: £24.8bn in lost output (ILC-UK 2014)
- Associated with depression, anxiety, dementia
- Good communication between patient and healthcare provider is essential for effective treatments across **all diseases**
 - unmanaged hearing loss significantly reduces the probability of this happening**

Figure 4.8: Mean EQ-5D scores for GP Patient Survey respondents with different sensory impairments by age, England, 2012-2013



Impact and total management of hearing loss

Hearing is fundamental to how we relate to the world and the people around us



Hearing devices can help:

- Restore lost function
- Manage difficult environments



But they cannot:

- Solve every problem
- Work all the time for everyone

Impact of hearing loss on the individual

Frustration & tension

Isolation

Withdrawal

Depression

Anxiety

Suicide



Hearing devices may not be sufficient
to mitigate these effects

Impact of hearing loss on the family

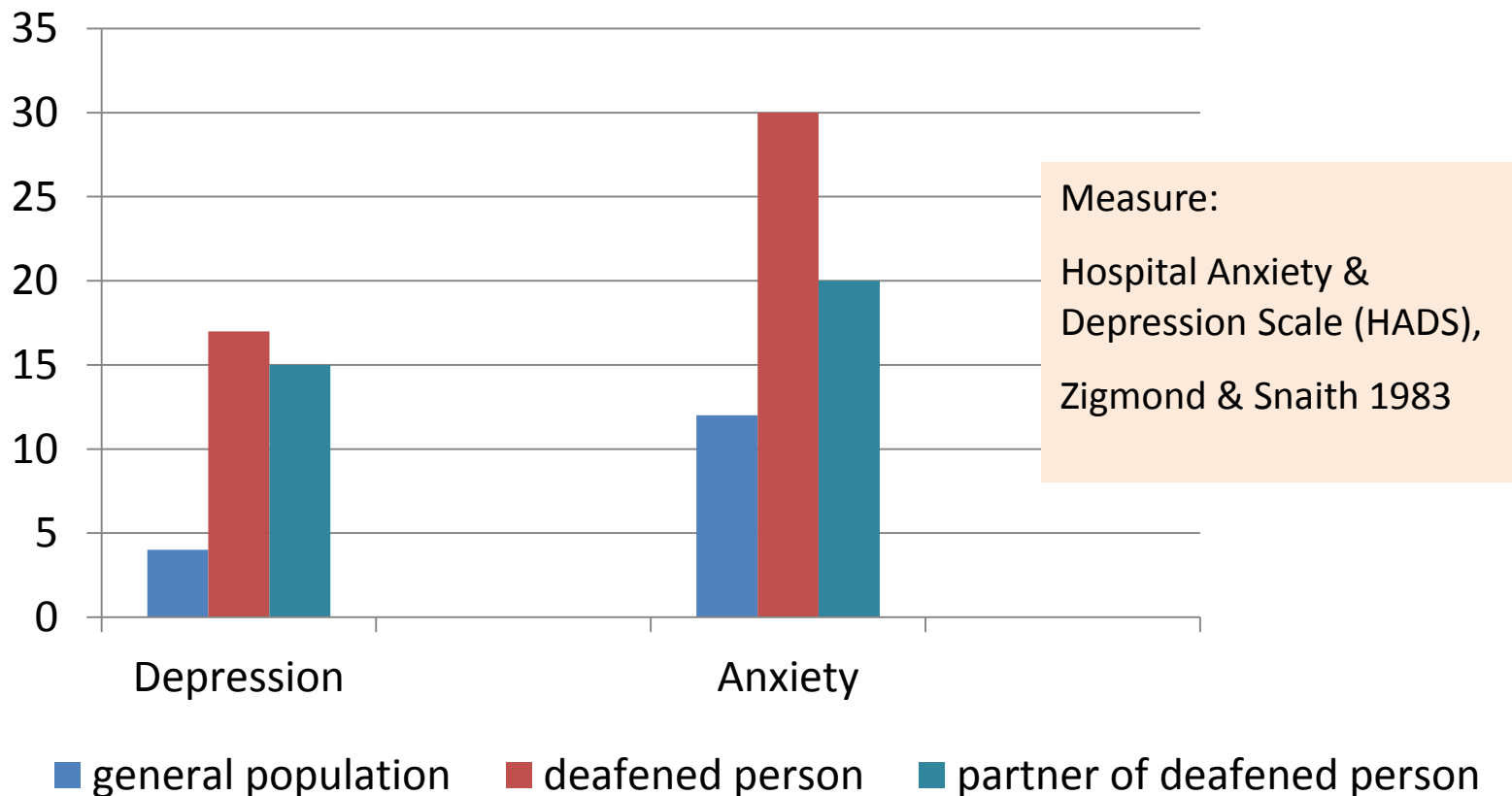
- Frustration & tension *in both partners*
- Isolation and withdrawal *for both partners*
- Depression & anxiety *in both partners*
- Family breakdown



Hearing devices may not be sufficient
to mitigate these effects

Mental health cost

Percent scoring above cut-off for clinical diagnosis



.Source: *Hidden Lives 2005* available from Hearing Link

Hearing loss in the workplace

Underemployment is a bigger problem than unemployment

- Increasing with demographic changes in the workplace
- Personal solutions are essential
- Reasonable adjustments are rarely made
- Access to Work often poorly applied
- The cost to society is significant
...and rising



Hearing devices may not be sufficient
to mitigate these effects

Hearing Link's support framework

Supports the individual and their family as they adjust to hearing loss



Delivered through peer support in the community



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